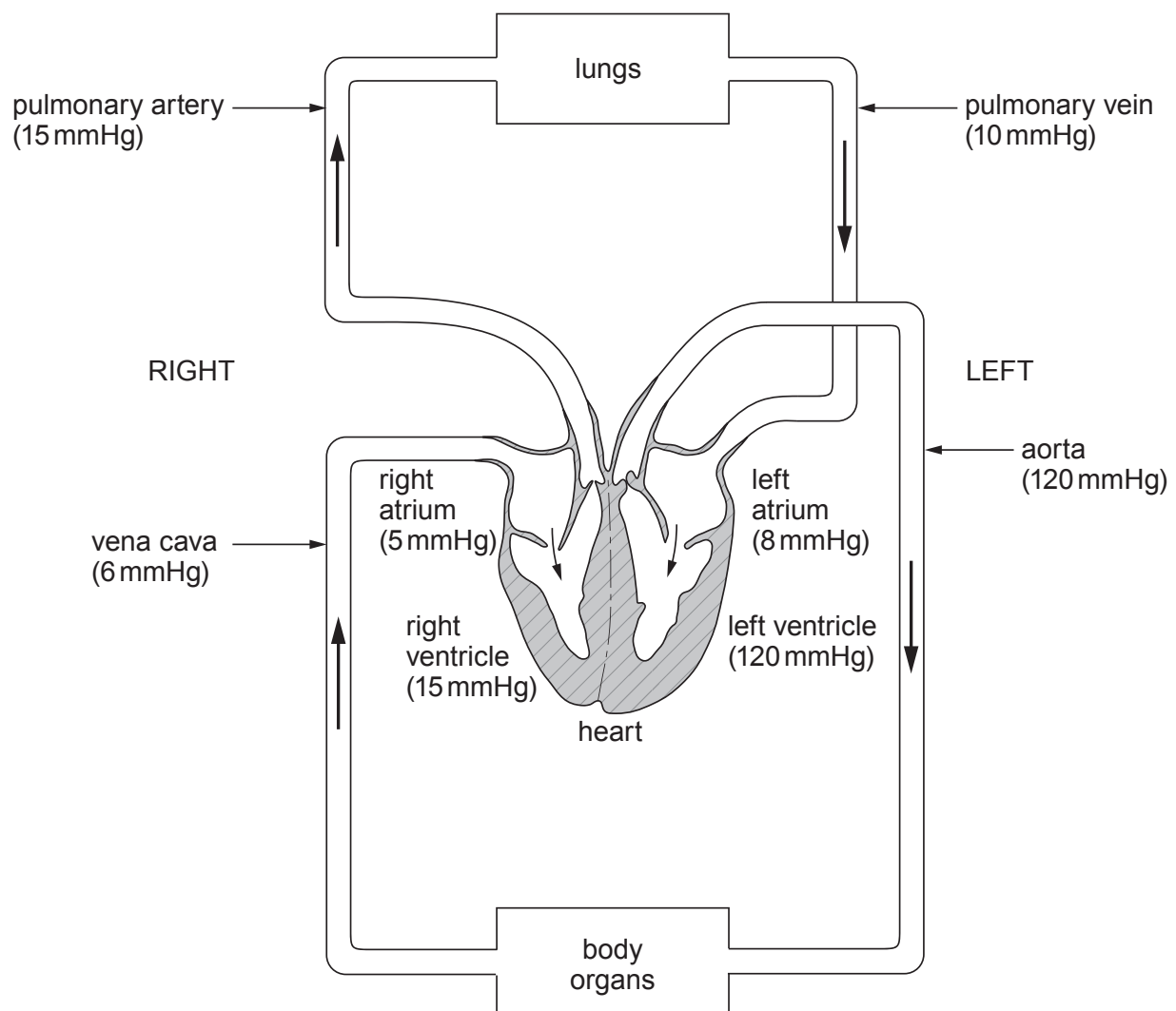


5. In one complete circulation of the body, blood passes through the heart twice. The blood travels:
- from the heart to the lungs and then back to the heart in the pulmonary circulation and then;
 - from the heart to the other organs of the body and back to the heart in the systemic circulation.

This double circulation is essential for the functioning of highly active animals such as mammals.

The diagram below represents the double circulation in humans. It shows the direction of blood flow and the pressure of the blood in various blood vessels in a healthy 25 year-old woman at rest.

The pressure of the blood is measured in mmHg (millimetres of mercury).



Examiner
only

(a) Suggest a reason for the difference in the:

(i) thickness of the walls of the atria and the ventricles; [1]

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(ii) blood pressure between the pulmonary artery and pulmonary vein; [1]

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(iii) blood pressure between the pulmonary artery and the aorta. [2]

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(b) Using the data for blood pressure in the **diagram opposite**, suggest why, after leaving the lungs, the blood has to return to the heart before it is sent to the other organs of the body. [2]

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(c) Describe how the direction of blood flow is maintained in the circulatory system. [1]

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7

